

A Combinatorial Model for Affine Demazure Crystals of Levels Zero and One

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Abstract. Crystals are colored directed graphs encoding information about Lie algebra representations. Certain affine crystals, known as Kirillov-Reshetikhin (KR) crystals, were realized in a uniform way (for all untwisted affine types) in terms of the quantum alcove model, and their graded characters were shown to coincide with the specialization of symmetric Macdonald polynomials at $t = 0$. In this paper we generalize these results by introducing a "non-symmetric" quantum alcove model for certain level zero Demazure-type crystals inside tensor products of KR crystals, whose graded characters coincide with non-symmetric Macdonald polynomials at $t = 0$. Moreover, in type A , we construct an explicit affine crystal isomorphism between the non-symmetric quantum alcove model and the semi-standard key tabloid model for level one affine Demazure crystals of type A by Assaf and González.

Keywords: Kirillov-Reshetikhin crystals, Demazure crystals, Macdonald polynomials, quantum alcove model, semistandard key tabloids.

1 Introduction

Certain representations of Lie algebras $\mathfrak{g}$ (including the highest weight ones) possess *crystal bases*, when viewed as representations of the corresponding quantum algebras $U_q(\mathfrak{g})$ [9]. In the limit $q \rightarrow 0$, the structure of a crystal basis is encoded in a colored directed graph on the basis elements, called a *crystal graph*. The directed i -edges give the action of the *Kashiwara operators* (analogues of the Chevalley generators) as $q \rightarrow 0$. Crystal graphs have various combinatorial models, and are useful tools in representation theory.

Kirillov–Reshetikhin (KR) modules [11] are finite-dimensional modules $W^{r,s}$ for affine Lie algebras, not of highest weight, labeled by a positive integer multiple $s\omega_r$ of the corresponding finite-type fundamental weight ω_r . We refer to the KR modules $W^{r,1}$ as single-column ones. In most cases, $W^{r,s}$ was shown to have a crystal basis, and the crystal graph is denoted $B^{r,s}$. The latter are building blocks for the highest weight crystals, constructed as infinite tensor products of certain $B^{r,s}$.

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In [14] it was shown that the *quantum alcove model*, constructed in [12], is a uniform model for tensor products of single-column KR crystals, for all untwisted affine types. The quantum alcove model generalizes the alcove model in [15], which describes the highest weight crystals of symmetrizable Kac-Moody algebras. These models are based on enumeration of certain paths in the quantum Bruhat graph (resp. the Bruhat graph) for the finite Weyl group.

In classical types there are (type-specific) models for tensor products of KR crystals $B^{r,1}$ based on fillings of Young diagrams consisting of concatenations of *Kashiwara-Nakashima (KN) columns* [10]. While the tableau models are simpler than the alcove models, they have less easily accessible information, so it is hard to use them in computations, for instance of charge, the energy function, and the combinatorial R -matrix (see 2.2). As these computations are much simpler in the quantum alcove model [14], an alternative is to relate them to the tableau models, via affine crystal isomorphisms from the former model to the latter one. Such maps, to be called “alcoves-to-fillings” maps, were constructed in types A, C in [12], and in types B, D in [4].

A corollary of the realization of tensor products of KR crystals in terms of the quantum alcove model is their connection with Macdonald polynomials (of the corresponding untwisted affine type). The symmetric Macdonald polynomials $P_\lambda(x; q, t)$, indexed by dominant weights λ , are Weyl group invariant polynomial with rational function coefficients in parameters q, t [17]. They generalize the irreducible characters of the corresponding simple Lie algebras, which are recovered for $q = t = 0$. A uniform combinatorial formula for $P_\lambda(x; q, t)$ was given by Ram and Yip [19] in terms of *alcove walks*. By comparing with the quantum alcove model, it was shown in [14] that the specialization $P_\lambda(x; q, t = 0)$ coincides with the graded character of a tensor product of single-column KR modules, graded is by the energy function.

There are also non-symmetric versions of Macdonald polynomials $E_\mu(x; q, t)$, where μ is an arbitrary weight [17], as well as a Ram-Yip formula for them. Specialized non-symmetric Macdonald polynomials are closely related to Demazure characters of affine Kac-Moody algebras, as discussed in [8]. Later, in [13], it was shown that $E_\mu(x; q, t = 0)$ coincides with the graded character of a Demazure-type submodule of a tensor product of single-column KR modules. As discussed in the mentioned paper, the latter can be thought of as a Demazure submodule of a certain quotient of a *level zero extremal weight module*. The corresponding crystal is called a *DARK crystal* (Kirillov–Reshetikhin Affine Demazure) in [3]. Thus, we expect a combinatorial model for the mentioned DARK crystal based on the alcove walks in the respective non-symmetric Ram-Yip formula specialized at $t = 0$. This is the first main result of the paper, and is a “non-symmetric version” of the quantum alcove model.

On the other hand, in type A , $E_\mu(x; q, t = 0)$ is expressed in terms of *semistandard key tabloids*. These were introduced in [1] as special cases of the *non-attacking skyline fillings* in the Haglund-Haiman-Loehr formula for non-symmetric Macdonald polynomials [6]. They

were given an affine crystal structure in [2], which realizes the crystals of level one Demazure modules for the affine Lie algebra $\widehat{\mathfrak{sl}}_n$. The second main result of our paper is the construction of an explicit affine crystal isomorphism between the non-symmetric quantum alcove model in type A and semistandard key tabloids, i.e., an alcoves-to-fillings map in the type A Demazure case.

This result is expected to generalize to the other classical types B, C , and D . We will construct alcoves-to-fillings correspondences which would extend the maps in [4, 12] mentioned above to the Demazure case. These constructions will combine those in [4, 12] with those in the current paper. In particular, we will derive the types B, C , and D analogues of the semistandard key tabloids, which will be Demazure-type analogues of the corresponding concatenations of KN columns.

2 Background

2.1 Root systems

Let $\mathfrak{g}$ be a complex simple Lie algebra, and $\mathfrak{h} \subset \mathfrak{g}$ a Cartan subalgebra. Let $\Phi \subset \mathfrak{h}_{\mathbb{R}}^*$ be the corresponding irreducible root system, and Φ^+ the set of positive roots, for which we use the notation $\alpha > 0 \Leftrightarrow \text{sgn}(\alpha) = 1$. Let $\alpha_i \in \Phi^+$ be the *simple roots* for i in some indexing set I and let θ be the highest root of. We denote by $\langle \cdot, \cdot \rangle$ the nondegenerate scalar product on $\mathfrak{h}_{\mathbb{R}}^*$ induced by the Killing form. Given a root α , we consider the corresponding coroot $\alpha^\vee := 2\alpha / \langle \alpha, \alpha \rangle$ and reflection s_α . The *weight lattice* P is generated by the *fundamental weights* ω_i for $i \in I$, which satisfy $\langle \omega_j, \alpha_i^\vee \rangle = \delta_{i,j}$. The set of dominant weights is denoted P^+ . Define $\rho := \sum_{i \in I} \omega_i$, and for a root α , set $|\alpha| := \text{sgn}(\alpha)\alpha$.

Let W be the corresponding finite Weyl group with Coxeter generators $s_i := s_{\alpha_i}$. With respect to these generators, W comes with a length function $\ell(\cdot)$. Let $w_\circ$ be the longest element of W . The Bruhat order on W is defined as the transitive closure of its covers: $w \lessdot ws_\alpha$ if $\ell(w) + 1 = \ell(ws_\alpha)$.

For a root $\alpha \in \Phi^+$ and an integer $k \in \mathbb{Z}$, consider the reflection $s_{\alpha,k}$ in the affine hyperplane $H_{\alpha,k} := \{\lambda \in \mathfrak{h}_{\mathbb{R}}^* \mid \langle \lambda, \alpha^\vee \rangle = k\}$. The *affine Weyl group* W_{aff} for the dual root system $\Phi^\vee$ is generated by such reflections. The hyperplanes $H_{\alpha,k}$ divide $\mathfrak{h}_{\mathbb{R}}^*$ into connected components called *alcoves*. The *fundamental alcove* is denoted $A_\circ$. An important object in defining the quantum alcove model is a certain directed graph on W .

Definition 1. The quantum Bruhat graph of W is the directed graph with vertex set W and edges $w \xrightarrow{\beta} ws_\beta$ labeled by $\beta \in \Phi^+$ if:

- (1) $w \lessdot ws_\beta$, or
- (2) $\ell(ws_\beta) = \ell(w) - 2\langle \beta^\vee, \rho \rangle + 1$.

2.2 Kirillov–Reshetikhin crystals

Given a semisimple or affine Lie algebra $\mathfrak{g}$ with simple roots α_i for $i \in I$, a $\mathfrak{g}$ -crystal is a nonempty set B together with maps $e_i, f_i : B \rightarrow B \sqcup \{0\}$ for $i \in I$ and $\text{wt} : B \rightarrow P$ satisfying $f_i(b) = b'$ if and only if $e_i(b') = b$ for $b \in B$. The maps e_i, f_i are called crystal operators and they give B the structure of a colored directed graph with edges $b \xrightarrow{i} f_i(b)$. For $\mathfrak{g}$ -crystals B_1 and B_2 , the tensor product $B_1 \otimes B_2$ is defined on the set $B_1 \times B_2$ with a rule that determines on which factor e_i and f_i act [7].

Kirillov–Reshetikhin (KR) modules, denoted $W^{r,s}$, are finite-dimensional modules, not of highest weight, of affine Lie algebras. They are indexed by positive integer multiples of a fundamental weight $s\omega_r$. In classical types $s\omega_r$ corresponds to a rectangle of height r and length s . In the case $s = 1$, the modules are called "single-column" KR modules. In most cases, $W^{r,s}$ was shown to have a crystal basis, and the corresponding *Kirillov–Reshetikhin crystal* is denoted $B^{r,s}$. For a composition $\mathbf{a} = (a_1, a_2, \dots, a_k)$, tensor products of column-shape KR crystals for untwisted types are defined as follows:

$$B = B^{\mathbf{a}} := \bigotimes_{i=1}^k B^{a_i, 1}. \quad (2.1)$$

The crystal B is known to be connected as an affine crystal, but disconnected as a classical crystal. If $\mathbf{a}'$ is a permutation of $\mathbf{a}$, then there is a unique crystal isomorphism between $B^{\mathbf{a}}$ and $B^{\mathbf{a}'}$, called the *combinatorial R-matrix*.

Let $\widehat{\mathfrak{g}}$ be an affine Lie algebra and let $\lambda \in P^+$ be a dominant weight for $\mathfrak{g}$. Consider the *level zero extremal weight module* $V(\lambda)$ over the quantum group $U_q(\widehat{\mathfrak{g}})$, generated by a vector v_λ of weight λ . For a Weyl group element w , the *Demazure submodule* $V_w(\lambda)$ is given by $V_w(\lambda) := U_q^+(\mathfrak{g}) \cdot wv_\lambda$.

Beck and Nakajima introduced a subtle finite-dimensional quotient of $V_{w_0}(\lambda)$, denoted $U_{w_0}^+(\lambda)$ (see [13, §3.3]). As a $U_q(\mathfrak{g})$ -module, $U_{w_0}^+(\lambda)$ is isomorphic to a tensor product of KR modules $W^{i,1}$. The image of the level zero Demazure module $V_w^+(\lambda)$ under the projection to $U_{w_0}^+(\lambda)$ is denoted $U_w^+(\lambda)$ and is a Demazure-type submodule.

2.3 The quantum alcove model

Fix a dominant weight $\lambda \in P^+$. The quantum alcove model depends on a sequence of roots $\Gamma := (\beta_1, \beta_2, \dots, \beta_m)$ called a λ -chain [12]. This is equivalent to a shortest sequence of adjacent alcoves from the fundamental alcove A_0 to the translate $A_0 - \lambda$. Define the reflection $r_i := s_{\beta_i}$.

Definition 2. A subset $J = \{j_1, j_2, \dots, j_s\} \subset [m]$ (possibly empty) is called *admissible* if we have a path in the quantum Bruhat graph

$$1 \rightarrow r_{j_1} \rightarrow r_{j_1} r_{j_2} \rightarrow \dots \rightarrow r_{j_1} r_{j_2} \dots r_{j_s} := \text{end}(J).$$

Denote by $\mathcal{A}(\Gamma)$ the set of admissible subsets of J .

Theorem 3. [12, 14] Let $\lambda = \omega_{i_1} + \omega_{i_2} + \cdots + \omega_{i_k}$ be a dominant weight of a untwisted affine Lie algebra. Then $\mathcal{A}(\Gamma)$ with the proper crystal operators e_i, f_i is a model for the tensor product of KR crystals $B^\lambda := B^{i_1,1} \otimes B^{i_2,1} \otimes \cdots \otimes B^{i_k,1}$.

Let $W_\lambda \subset W$ be the stabilizer of λ . Given a Weyl group element u , we denote by $[u]$ the lowest coset representative of uW_λ .

Theorem 4. [13] The set of admissible subsets $J \in \mathcal{A}(\Gamma)$ satisfying the condition $\text{end}(J) \leq [w]$ is a model for the DARK crystal corresponding to the Demazure-type module $U_w^+(\lambda)$.

Based on the quantum alcove model, the following is derived in [13, 14]:

Theorem 5. Let $\hat{\mathfrak{g}}$ be an untwisted affine Lie algebra and $\lambda \in P^+$ a dominant weight of $\mathfrak{g}$.

1. The symmetric Macdonald polynomial $P_\lambda(x; q, t = 0)$ is a graded character of the tensor product of column-shape KR crystals B^λ , so it is expressed by $\mathcal{A}(\Gamma)$.
2. The nonsymmetric Macdonald polynomial $E_{w\lambda}(x; q, t = 0)$ is a graded character of the Demazure-type crystal corresponding to $U_w^+(\lambda)$, so it is expressed by the subset of $\mathcal{A}(\Gamma)$ defined in Theorem 4.

2.4 Semistandard key tabloids and their crystal structure

In this section we describe the model defined in [1] for the level one affine Demazure crystals in type A_{n-1} . The objects are fillings of diagrams subject to conditions on pairs and triples of entries. A *diagram* $\text{dg}(\mu)$ of the weak composition $\mu = (\mu_1, \mu_2, \dots, \mu_n)$ consists of μ_i boxes left-justified in row i . We label the box in row i column j as (i, j) . For

example, if $\mu = (3, 0, 2)$ then $\text{dg}(\mu) =$

$$\begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array}.$$

Definition 6. A filling is an assignment of a positive integer $m_{(i,j)} \leq n$ to each box (i, j) .

Definition 7. Given a diagram $\text{dg}(\mu)$, we say two boxes are attacking if they are in the same column or they are in adjacent columns with the left box strictly higher than the right box. A filling is non-attacking if every pair of attacking boxes have distinct entries and no entry in the first column exceeds its row index.

Definition 8. A triple is a collection of three boxes with two row adjacent and either (Type I) the third cell is above the left and the lower row is strictly longer, or (Type II) the third cell is below the right and the higher row is weakly longer. The orientation of a triple is determined by reading the entries of the boxes from smallest to largest. A coinversion triple is a Type I triple oriented counterclockwise or a Type II triple oriented clockwise.

Definition 9. [2] The semi-standard key tabloids of shape μ , denoted $\text{SSKD}(\mu)$, are the non-attacking fillings of $\text{dg}(\mu)$ with no coinversion triples.

Example 10. The filling $\begin{array}{|c|c|} \hline 3 & 2 \\ \hline 1 & 1 & 3 \\ \hline \end{array}$ is a semi-standard key tabloid of shape $\mu = (3, 0, 2)$.

The *weight* of a tabloid T , denoted $\text{wt}(T)$ is defined as the weak composition with i -th part equal to the number of entries i in T . For example $\text{wt}(T) = (2, 1, 2)$ in the example above. Assaf and González defined raising and lowering operators e_i, f_i for $i = 0, 1, \dots, n-1$, which act on T by swapping certain entries i and $i+1$ (1 and n for $i = 0$). The affected entries are determined by a pairing rule [2, defs 3.1-3.6]. The operators e_i, f_i , and the map wt define a crystal structure on $\text{SSKD}(\mu)$.

Theorem 11. [2] The above crystal structure on semi-standard key tabloids of shape μ is isomorphic to a certain level one Demazure crystal of $\widehat{\mathfrak{sl}}_n$.

Theorem 12. [6] The type A specialized non-symmetric Macdonald polynomial is given by

$$E_\mu(x; q, t = 0) = \sum_{T \in \text{SSKD}(\mu)} q^{\text{maj}(T)} x^{\text{wt}(T)}, \quad (2.2)$$

where $\text{maj}(T)$ is the sum of all legs of boxes (i, j) of T with $m_{(i,j)} < m_{(i,j+1)}$.

Remark 13. The level zero and level one Demazure crystals discussed above are known to be isomorphic when the affine root α_0 is short [5].

3 Nonsymmetric quantum alcove model

We introduce a new model for the crystal of the Demazure-type module $U_w^+(\lambda)$. The objects of the model are non-symmetric analogues of admissible subsets as in Definition 2. This model has two benefits over the original quantum alcove model: 1) of removing the condition $\text{end}(J) \leq \lfloor w \rfloor$ in Theorem 4, and 2) of being compatible with a tableau model for a Type A level one affine Demazure crystal, to be discussed in Section 4.

3.1 w -admissible subsets

Given an arbitrary weight μ , let $w \in W$ be the element of maximal length such that $\mu = w\lambda$ for a dominant weight λ . Consider a reduced alcove path Γ from $A_\circ$ to $ww_\circ A_\circ + \mu$:

$$\Gamma := A_\circ \xrightarrow{-\gamma_1} A_1 \xrightarrow{-\gamma_2} A_2 \xrightarrow{-\gamma_3} \dots \xrightarrow{-\gamma_m} A_m = ww_\circ A_\circ + \mu. \quad (3.1)$$

The sequence of crossed hyperplanes is given by $H_{|\gamma_k|, m_k}$ where m_k is defined by this relation. To describe the objects of the model, we need another alcove path Γ' obtained from Γ by applying $-w^{-1}$ on the left:

$$\Gamma' = -w^{-1}\Gamma = -w^{-1}A_o \xrightarrow{-\beta_1} A'_1 \xrightarrow{-\beta_2} A'_2 \xrightarrow{-\beta_3} \dots \xrightarrow{-\beta_m} A'_m = A_o - \lambda. \quad (3.2)$$

Note that in this case all the roots β_i are positive. Let $J = \{j_1 < j_2 < \dots < j_s\} \subseteq [m]$ and define the reflections $r_{j_i} := s_{\gamma_{j_i}}$ for $i = 1, 2, \dots, s$. We can interpret J as a "folding" of the alcove path Γ , resulting in a "folded" alcove path $\Gamma(J) := (\delta_1, \delta_2, \dots, \delta_m)$, where

$$\delta_k = r_{j_1} r_{j_2} \dots r_{j_i}(\gamma_k). \quad (3.3)$$

Here $i = i(k)$ is the largest index for which $j_i < k$. After folding Γ , the k -th hyperplane crossed in the path $\Gamma(J)$ is given by:

$$H_{|\delta_k|, l_k^J} := \tilde{r}_{j_1} \dots \tilde{r}_{j_i} H_{|\gamma_k|, m_k}, \quad (3.4)$$

where $\tilde{r}_k := s_{\gamma_k, m_k}$ and l_k^J is defined by this relation. The objects of our model are subsets $J \subset [m]$ satisfying the following condition:

Definition 14. We say $J = \{j_1 < j_2 < \dots < j_s\} \subset [m]$ is w -admissible and write $J \in \mathcal{A}(\Gamma')$ if there is a path ending at w in the quantum Bruhat graph:

$$ws_{\beta_{j_1}} s_{\beta_{j_2}} \dots s_{\beta_{j_s}} \rightarrow \dots \rightarrow ws_{\beta_{j_1}} s_{\beta_{j_2}} \rightarrow ws_{\beta_{j_1}} \rightarrow w.$$

Define $\text{wt}(J) := \tilde{r}_{j_1} \tilde{r}_{j_2} \dots \tilde{r}_{j_s}(\mu)$ and $\text{ht}(J) := \sum_i l_{j_i}^J$ where the sum is over all i such that the edge corresponding to β_{j_i} is a quantum edge, i.e an edge of type (2) in Definition 1.

Theorem 15. [19, consequence of Ram-Yip formula]

$$E_\mu(x, q, 0) = \sum_{J \in \mathcal{A}(\Gamma)} q^{\text{ht}(J)} x^{\text{wt}(J)}. \quad (3.5)$$

3.2 Crystal operators and the first main result

Using the setup in the previous section, we define crystal operators e_i, f_i on $\mathcal{A}(\Gamma')$. For J a w -admissible subset, define:

$$\begin{aligned} I_\alpha(J) &:= \{i \in [m] \mid \delta_i = \pm \alpha\}, & \widehat{I}_\alpha &= \widehat{I}_\alpha(J) := I_\alpha \cup \{\infty\}, \\ l_\alpha^\infty &:= \langle \text{wt}(J), \text{sgn}(\alpha) \alpha^\vee \rangle, & \delta_\infty &:= r_{j_1} r_{j_2} \dots r_{j_s} v \rho. \end{aligned}$$

The following graphical representation of the levels l_i^J of the hyperplanes crossed by the folded path $\Gamma(J)$ is used in defining the crystal operators. Set:

$$\widehat{I}_\alpha = \{i_1 < i_2 < \dots < i_n < i_{n+1} = \infty\} \text{ and } \varepsilon_i := \begin{cases} 1 & \text{if } i \notin J \\ -1 & \text{if } i \in J \end{cases}.$$

If $\alpha > 0$, we define the continuous piece-wise linear function $g_\alpha : [0, n + \frac{1}{2}] \rightarrow \mathbb{R}$ by

$$g_\alpha(0) = -\frac{1}{2}, \quad g'_\alpha(x) = \begin{cases} \text{sgn}(\delta_{i_k}) & \text{if } x \in (k-1, k - \frac{1}{2}), k = 1, \dots, n \\ \varepsilon_{i_k} \text{sgn}(\delta_{i_k}) & \text{if } x \in (k - \frac{1}{2}, k), k = 1, \dots, n \\ \text{sgn}(\langle \delta_\infty, -\alpha^\vee \rangle) & \text{if } x \in (n, n + \frac{1}{2}). \end{cases} \quad (3.6)$$

If $\alpha < 0$, we define g_α to be the graph obtained by reflecting $g_{-\alpha}$ across the x -axis. For any α we have

$$\text{sgn}(\alpha) l_{i_k}^J = -g_\alpha \left(k - \frac{1}{2} \right) \text{ for } k = 1, \dots, n \quad (3.7)$$

$$\text{sgn}(\alpha) l_\alpha^\infty := \langle \text{wt}(J), -\alpha^\vee \rangle = -g_\alpha \left(n + \frac{1}{2} \right). \quad (3.8)$$

Let J be a w -admissible subset. Fix $i \in \{0, \dots, n-1\}$. If $i > 0$ then α_i is a simple root, and if $i = 0$ then we set $\alpha_0 := -\theta$ for the highest root θ .

Define $M := \max(g_{\alpha_i})$. Assuming that $M > \langle \text{wt}(J), \alpha_i^\vee \rangle$, let k be the maximum index j in I_{α_i} for which we have $\text{sgn}(\alpha_i) l_j^J = -M$, and let m be the successor of k in $\widehat{I}_{\alpha_i}$. Assuming also that $M \geq \delta_{i,0}$, we have $k \in J$, and either $m \notin J$ or $m = \infty$. Define:

$$f_i(J) := \begin{cases} (J \setminus \{k\}) \cup \{m\} & \text{if } M > \langle \mu(J), \alpha_i^\vee \rangle \text{ and } M \geq \delta_{i,0} \\ \mathbf{0} & \text{otherwise.} \end{cases} \quad (3.9)$$

The operators e_i are defined similarly. We use the convention that $J \setminus \{\infty\} = J \cup \{\infty\} = J$. Using the properties of the quantum Bruhat graph, we show that the set $\mathcal{A}(\Gamma')$ is closed under the crystal operators f_i, e_i , so the model is well-defined. Based on this, and by using a similar approach to the one in [13, 14], we derive our first main result.

Theorem 16. *The set of w -admissible subsets $\mathcal{A}(\Gamma')$ together with the operators e_i, f_i and map wt defined above is a model for the Demazure-type crystal corresponding to the module $U_w^+(\lambda)$.*

Remark 17. The entire crystal is constructed by repeated application of the operators e_i and f_i , starting from the lowest element $J = \emptyset$ ($f_i(J) = \mathbf{0}$ for all i). By comparison, the Demazure-type crystal from Theorem 4 is constructed by repeated application of only the operators f_i , starting from the highest element $J = \emptyset$ ($e_i(J) = \mathbf{0}$ for all i) but checking the condition $\text{end}(J) \leq \lfloor w \rfloor$ at each step.

Remark 18. Viewing J as a folding of the alcove path Γ , the crystal operator f_i acts by removing the fold along the hyperplane $H_{\alpha_i, -x}$ for $x \geq 0$ maximal, and adding a fold along the hyperplane $H_{\alpha_1, -(x+1)}$ (if such x exists). The action of e_i is similar, this time moving folds in one level.

4 Specializing the model to type A

4.1 A choice of μ -chain

Recall that the objects of the quantum alcove model depend on a choice of alcove path (different choices with yield isomorphic models). In order to obtain explicit maps from the quantum alcove model to the tabloid model, a particular alcove path Γ' is needed.

To this end, consider the setup as in Section 3, now specialized to type A_{n-1} . Let λ' be the conjugate partition to λ and let $\text{dg}(\mu)$ be an empty diagram of shape μ . Γ' will be constructed by concatenating smaller sequences of roots $\Gamma'(j)$, one for each column of $\text{dg}(\mu)$, i.e.

$$\Gamma' = \Gamma'(1)\Gamma'(2) \cdots \Gamma'(\lambda_1).$$

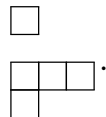
Suppose $\lambda'_j = k$. Rename and reorder $w(1), w(2), \dots, w(k)$ as $i_1, i_2, \dots, i_k$ with

$$i_1 < i_2 < \cdots < i_k. \quad (4.1)$$

For $s \leq k$, define $m_s := \min(\{a \in \{k+1, \dots, \lambda'_{j-1}\} \mid i_s > w(a)\} \cup \{\lambda'_{j-1} + 1\})$ (by convention we set $\lambda'_0 = n$). Then $\Gamma'(j)$ is given by the following sequence of roots:

$$\Gamma'(j) = \begin{pmatrix} (w^{-1}(i_1), m_1) & (w^{-1}(i_1), m_1 + 1) & \cdots & (w^{-1}(i_1), n) \\ \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots \\ (w^{-1}(i_k), m_k) & (w^{-1}(i_k), m_k + 1) & \cdots & (w^{-1}(i_k), n) \end{pmatrix}. \quad (4.2)$$

Proposition 19. [18] Γ' , defined as the above concatenation, is a sequence of positive roots $(\beta_1, \beta_2, \beta_3, \dots)$ encoding an alcove path of the form (3.2), with which it is identified.

Example 20. Let $\mu = (1, 0, 3, 1)$. Then $w = w^{-1} = 3412$, $\text{dg}(\mu) =$ .

The first column has 3 boxes, so $i_1 = 1, m_1 = 5$; $i_2 = 3, m_2 = 4$; and $i_3 = 4, m_3 = 4$.

The second column has 1 box, so $i_1 = 3, m_1 = 3$.

The third column has 1 box, so $i_1 = 3, m_1 = 2$.

The corresponding sequences of roots $\Gamma'(j)$ are:

$$\Gamma'(1) = ((1, 4)(2, 4)); \quad \Gamma'(2) = ((1, 3)(1, 4)); \quad \Gamma'(3) = ((1, 2)(1, 3)(1, 4)).$$

Concatenating gives the alcove path

$$\Gamma' = ((1,4)(2,4)|(1,3)(1,4)|(1,2)(1,3)(1,4)).$$

4.2 The filling map

We now describe a map from w -admissible subsets $\mathcal{A}(\Gamma')$ to semi-standard key tabloids $SSKD(\mu)$. It is a forgetful map that produces a filling from a subset of roots J .

Recall that a w -admissible subset selects a set of positions within the alcove path Γ' . Let T_j be the sequence of transpositions corresponding to the positions in $\Gamma'(j)$ that are selected, noting that T_j could possibly be empty. Given a w -admissible subset J and a column j of $\text{dg}(\mu)$, define:

$$u^j := wT_1T_2 \cdots T_j$$

where the right hand side is the permutation obtained from w by consecutively multiplying by each transposition present in $T_1, T_2, \dots, T_j$ from left to right.

Definition 21. The map *fill* is a map from subsets J (not necessarily w -admissible) to fillings of shape μ with entry in row i_a , column j given by

$$\text{fill}(J)(i_s, j) := u^j w^{-1}(i_s).$$

Theorem 22. If J is w -admissible, then $\text{fill}(J)$ is non-attacking and has no coinversion triples. Moreover, *fill* is a weight preserving and height preserving bijection between w -admissible subsets and semi-standard key tabloids of shape μ .

Example 23. Continuing Example 20, let $J = \{2, 5, 6, 7\}$. One can check that J is w -admissible.

$$T_1 = (2, 4); \quad T_2 = \emptyset; \quad T_3 = (1, 2)(1, 3)(1, 4).$$

Starting from w , we apply in order the transpositions above:

$$w = \begin{array}{|c|} \hline 2 \\ \hline 1 \\ \hline 4 \\ \hline 3 \\ \hline \end{array} \xrightarrow{(2,4)} \begin{array}{|c|} \hline 4 \\ \hline 1 \\ \hline 2 \\ \hline 3 \\ \hline \end{array} \quad \bigg| \quad \begin{array}{|c|} \hline 4 \\ \hline 1 \\ \hline 2 \\ \hline 3 \\ \hline \end{array} \xrightarrow{(1,2)} \begin{array}{|c|} \hline 4 \\ \hline 1 \\ \hline 3 \\ \hline 2 \\ \hline \end{array} \xrightarrow{(1,3)} \begin{array}{|c|} \hline 4 \\ \hline 2 \\ \hline 3 \\ \hline 1 \\ \hline \end{array} \xrightarrow{(1,4)} \begin{array}{|c|} \hline 1 \\ \hline 2 \\ \hline 3 \\ \hline 4 \\ \hline \end{array}$$

$$u^1 = 3412(2, 4) = \mathbf{3214}; \quad u^2 = \mathbf{3214}; \quad u^3 = 3214(1, 3)(1, 4) = \mathbf{4231}$$

Recall that $w^{-1} = 3412$. The entry in box (i_s, j) is $u^j(w^{-1}(i_s))$, so $\text{fill}(J) =$

$$\begin{array}{|c|c|c|} \hline 2 & & \\ \hline 3 & 3 & 4 \\ \hline \end{array}$$

$$\begin{array}{|c|} \hline 1 \\ \hline \end{array}$$

4.3 The inverse map

The existence of a map from semi-standard key tabloids to w -admissible subsets depends on the following order that is closely related to the quantum Bruhat graph for $W = S_n$.

Definition 24. Define an order $\prec_i$ on $[n]$ by:

$$i \prec_i i+1 \prec_i i+2 \prec_i \cdots \prec_i n \prec_i 1 \prec_i \cdots \prec_i i-1.$$

To reconstruct the w -admissible subset J from a semi-standard key tabloid T such that $\text{fill}(J) = T$, a “greedy” algorithm is run once for each box of T , working up the columns then across the rows. The initial input is the permutation w . If the target box is in column j , row i_a and has entry b , we select every root of $\Gamma'(J)$ (4.2) of the form $(w^{-1}(i_a), m_a + x)$ (viewed as a transposition) that brings us closer to our target entry (with respect to the order $\prec_b$) and consecutively multiply the current permutation by each selected transposition.

Proposition 25. Application of the process described above terminates, produces a unique path in the quantum Bruhat graph, and is the inverse of the map fill .

4.4 The second main result

Here we recap the main constructions and state our main result. For a dominant weight λ and a Weyl group element w , we construct a crystal structure on w -admissible subsets that realizes the crystal of the level zero Demazure-type module $U_w^+(\lambda)$. In type A , Assaf and González constructed a crystal on semi-standard key tabloids that realizes a certain level one affine Demazure crystal. Through our bijection fill , we show that the crystal operators on tabloids are compatible with our crystal operators on w -admissible subsets. Moreover, fill preserves the statistics ht and maj , which compute the *charge* statistic [16].

Theorem 26. The map fill is a crystal isomorphism from w -admissible subsets to semi-standard key tabloids. Moreover, $\text{ht}(J) = \text{maj}(\text{fill}(J))$.

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